Discrete and Continuous Distributions

50 minutes • 50 points • Student quiz

Name: _____ **Date:** _____

Instructions. Answer all four problems and show your reasoning. No calculator is needed; exact expressions with powers, binomial coefficients, or e are acceptable. You may quote standard distribution formulas. In Problem 4, build the answer from the stages instead of simply quoting a coupon-collector formula. Geometric variables count trials *including the first success*; exponential parameters are *rates*. The suggested times leave 4 minutes for checking.

1. One trial, a count, and a wait

12 points · about 11 minutes

A device makes independent attempts to send a message. Each attempt succeeds with probability $1/3$.

(a) Let $I = 1$ if the first attempt succeeds and $I = 0$ otherwise. Name the distribution of I, including its parameter, and find $\mathbb{E}[I]$ and $\mathrm{Var}(I)$. (3 points)

(b) Let N be the number of successes in the first six attempts. Name its distribution and parameters. Find $\mathbb{P}(N = 2)$, $\mathbb{E}[N]$, and $\mathrm{Var}(N)$. (5 points)

(c) Let T be the number of attempts up to and including the first success. Name its distribution and parameter, state its possible values, and find $\mathbb{P}(T > 3)$ and $\mathbb{E}[T]$. (4 points)

2. Counts and waiting times

10 points · about 9 minutes

A service receives requests according to this model: the number arriving in any interval of length t minutes has a Poisson distribution with parameter $2t$, and counts in disjoint intervals are independent. Let N be the number of requests in the next *30 seconds*, and let W be the waiting time, in *minutes*, from now until the next request.

(a) State the distribution and parameter of N, and find $\mathbb{E}[N]$ and $\mathrm{Var}(N)$. (3 points)

(b) Find $\mathbb{P}(N \geq 2)$. (2 points)

(c) Use the probability of *zero requests* in the next t minutes to find $\mathbb{P}(W > t)$ for $t \geq 0$. Identify the distribution of W, including its rate, and find $\mathbb{E}[W]$. (3 points)

(d) Find $\mathbb{P}(W > 3 \mid W > 2)$ and justify your answer by a calculation or the memoryless property. (2 points)

3. A continuous uniform wait

8 points · about 8 minutes

For a different service, the waiting time U, in minutes, has a *continuous* uniform distribution on $[0, 6]$.

(a) Write the density $f_U(u)$, including its value outside $[0, 6]$. Find $\mathbb{P}(2 < U < 5)$ and $\mathbb{P}(U = 3)$. (3 points)

(b) Find $\mathbb{E}[U]$ and $\mathrm{Var}(U)$. (2 points)

(c) Is this uniform waiting time memoryless? Compute $\mathbb{P}(U > 5 \mid U > 2)$ and $\mathbb{P}(U > 3)$, then compare them. (3 points)

4. Coupon collection, one stage at a time

20 points · about 18 minutes

Each pack contains one sticker, equally likely to be of type A, B, C, or D. Packs are independent; duplicates are possible. You start with no stickers. Let C be the number of packs needed to collect all four types. After exactly k distinct types have been collected, let W_k count *additional packs up to and including the next new type*, for $k = 0, 1, 2, 3$.

You may use: if $G \sim \text{Geom}(p)$ counts trials through the first success, then $\mathbb{E}[G] = 1/p$.

(a) Your first five stickers are A, A, C, A, C. Which types would make the next sticker new, and what is the probability it is new? (2 points)

(b) After k distinct types have been collected, find the probability p_k that the next pack gives a new type. Why does this probability stay the same when a duplicate appears? (3 points)

(c) State the distribution of W_k for $k = 1, 2, 3$. Fill in the table, treating W_0 directly. (5 points)

Distinct types k	New-type probability p_k	Expected wait $\mathbb{E}[W_k]$
0	_____	_____
1	_____	_____
2	_____	_____
3	_____	_____

(d) Express C in terms of W_0, W_1, W_2, W_3 and compute $\mathbb{E}[C]$. Name the expectation rule you use. Does that rule require the stages to be independent? (4 points)

(e) Suppose you currently have exactly two distinct types. What is the expected number of *additional* packs to finish? Explain why $4/2 = 2$ is not the answer to this question. (3 points)

(f) A student claims that C itself has a geometric distribution with parameter $1/4$. Is this correct? Give a reason using the possible values of C or the changes between stages. (3 points)